Behavioral Stability of Quantized Large Language Models Under Prompt Drift: The Resilio Evaluation Framework

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***Abstract*—Large Language Models are increasingly deployed using quantization techniques such as 8-bit and 4-bit precision to reduce memory footprint and inference costs. While these models perform well on clean benchmarks, real-world deployment encounters "Prompt Drift"—typographical errors, informal phrasing, and structural degradation.**

This paper introduces Resilio, a systematic evaluation framework investigating the interaction between model quantization and five progressive levels of prompt quality degradation across Llama 3.1 8B, Mistral 7B, and Phi-3 Mini. We evaluate behavioral stability using two metrics: Task Performance Score (TPS) and Behavioral Stability Score (BSS).

Our comprehensive study of 5,760 model responses reveals that modern quantization methods introduce minimal additional degradation beyond the primary drift effect. Across all tested configurations, 4-bit quantization adds only 1–3% stability loss compared to FP16 baselines, while prompt drift alone causes 15–35% degradation. However, task-specific patterns emerge: reasoning-intensive tasks show marginally higher quantization sensitivity than classification tasks.

These findings challenge assumptions about quantized model fragility and suggest that deployment decisions should prioritize model architecture selection and input quality control over quantization level choice for most practical applications.

***Index Terms*—Quantization, Large Language Models, Prompt Drift, Behavioral Stability, BSS, TPS, Robustness, Deployment**

I. Introduction

The deployment of Large Language Models has shifted from centralized data centers to resource-constrained edge environments, driven by demands for reduced latency, enhanced privacy, and lower operational costs. Post-training quantization has become the standard optimization strategy, compressing model weights from 16-bit floating point (FP16) to 8-bit or 4-bit precision. While these methods achieve substantial memory compression with minimal impact on clean-input accuracy, their behavior under realistic input degradation remains poorly characterized.

A. The Phenomenon of Prompt Drift

Standard evaluation protocols assess LLMs using "ideal" benchmarks characterized by curated, well-structured syntax. However, production environments expose models to a continuous spectrum of input degradation—a phenomenon we formalize as *Prompt Drift*. Prompt Drift reflects the naturally occurring noise in human-AI interaction, spanning five escalating severity levels:

- **Typographical Noise:** Character-level adjacency errors and transpositions.
- **Formatting Loss:** Omission of capitalization and punctuation.
- **Linguistic Informality:** Use of slang, abbreviations, and colloquial language.
- **Structural Breakdown:** Grammatical inconsistencies and sentence corruption.

B. The Critical Evaluation Gap

A fundamental question exists in deployed AI: how does quantization interact with realistic input degradation? Prior work evaluates quantization on clean benchmarks and robustness on full-precision models, leaving their interaction largely unmeasured. Two competing hypotheses exist: quantization may amplify sensitivity to input noise through reduced numerical precision, or modern compression techniques may prove robust enough that quantization effects are negligible compared to drift effects.

The *Resilio* framework addresses this question through systematic evaluation across model families, quantization levels, task categories, and drift severities. By implementing multi-dimensional stability scoring, we establish empirical measurements of the quantization-drift interaction to inform deployment decisions.

II. Related Work

The pursuit of deploying Large Language Models (LLMs) on edge devices has catalyzed research into two previously distinct domains: computational optimization through quantization and the linguistic robustness of models under perturbation. The *Resilio* framework operates at the intersection of these fields.

A. LLM Quantization for Efficient Deployment

The necessity for on-device LLM execution has led to the rapid maturation of Post-Training Quantization (PTQ) techniques. As noted by Dettmers *et al.* [1], 8-bit matrix multiplication allows for significant memory reduction with negligible performance loss on curated benchmarks. The introduction of 4-bit quantization further pushes these boundaries, enabling models like Phi-3, Llama 3.1, and Mistral 7B to operate within the strict memory constraints of mobile hardware [2], [3], [19].

However, while prior analyses provide extensive evaluation of the resource efficiency of INT4 versus INT8 representations [14], [18], these studies typically rely on "clean" input distributions. Recent work suggests that such extreme compression levels may introduce hidden instabilities or alter model safety alignment—a phenomenon sometimes termed *quantization-based vulnerability* [10], [13].

B. Robustness to Input Perturbations and Prompt Drift

The sensitivity of LLMs to surface-level changes in input text is a well-documented challenge. Research into typographical noise [12] and naturally occurring errors in training data [9] confirms that even slight character-level deviations can produce significant variance in model output. Frameworks such as PromptBench have begun to standardize the measurement of model resilience against adversarial attacks [5], [21].

Furthermore, studies on "noisy instructions" indicate that the quality of the prompt is a primary determinant of zero-shot generalization [11]. Despite these insights, existing robustness studies primarily utilize full-precision (FP16 or FP32) models, leaving the stability of quantized architectures under similar *Prompt Drift* largely unmeasured [4].

C. Behavioral Stability and Runtime Monitoring

As AI agents move into production, *behavioral stability* has emerged as a more critical metric than simple accuracy. This stability can be defined as the consistency of model behavior across intra-prompt variations [6]. In real-world deployments, *data drift*—the shift in the distribution of user inputs over time—presents a constant risk to model reliability [7].

While the research community has proposed various runtime monitoring strategies to detect behavioral shifts in AI systems [20], there remains a lack of unified metrics that measure stability relative to a clean baseline. The interaction between acceleration techniques and these hidden instabilities remains an open area of inquiry [13].

D. Identification of Research Gaps

A systematic review of the literature reveals a critical tripartite gap:

- **Isolation of Factors:** Quantization is typically evaluated on clean inputs, while robustness is tested on full-precision models. The interaction between bit-precision and input drift is largely omitted.
- **Absence of Graduated Drift Frameworks:** Most studies rely on binary noise conditions (clean vs. noisy) rather than a structured, multi-level taxonomy (L1–L5) as proposed in this work.
- **Metric Limitations:** Standard accuracy metrics fail to capture *silent failures*, where a model maintains high confidence despite producing logically inconsistent or incorrect outputs.

By introducing the *Task Performance Score (TPS)* and *Behavioral Stability Score (BSS)*, the *Resilio* framework provides a multi-factor analysis of task-specific stability patterns under quantized prompt drift.

III. The Resilio Methodology

The architecture of the *Resilio* framework is centered on a modular, reproducible pipeline that bridges the gap between synthetic benchmarks and the stochastic nature of production-level input noise. The methodology is executed through three distinct phases: custom dataset curation, object-oriented drift simulation, and semantic validation.

A. Dataset Architecture

To facilitate a granular analysis of model stability, we curated a specialized dataset comprising 40 base prompts. While these prompts are strategically categorized to mirror established linguistic challenges, they were custom-developed to target specific behavioral boundaries in Large Language Models:

- **Reasoning:** 10 prompts engineered to evaluate multi-step mathematical logic and sequential deduction.
- **Logical Inference:** 10 prompts structured to assess complex deductive reasoning and syllogistic consistency.
- **Question Answering:** 10 prompts focused on the precision of factual retrieval and context-window utilization.
- **Sentiment Classification:** 10 prompts designed to measure the robustness of pattern recognition under lexical noise.

B. OOP-based Drift Generation Framework

The technical core of *Resilio* is an Object-Oriented Programming (OOP) based drift generator. This modular engine allows for the application of graduated, additive transformations that simulate realistic user-driven degradation. Each base prompt was subjected to 16 unique variations to ensure statistical diversity in the results.

The framework formalizes *Prompt Drift* into a five-tier severity taxonomy:

- **L1 (Character-level Perturbation):** Simulates typographical errors via keyboard adjacency mapping and character transpositions.
- **L2 (Form Degradation):** Implements systematic removal of casing, standard punctuation, and formal whitespace.
- **L3 (Linguistic Shift):** Replaces formal lexical units with slang, non-standard abbreviations, and colloquial expressions.

- **L4 (Grammatical Breakdown):** Induces word merging, omission of essential function words, and overall syntactic collapse.
- **L5 (Structural Truncation):** Combines aggressive informality with content reduction, forcing the model to infer intent from fragmented inputs.

C. Mathematical Validation of Realism

To ensure that the generated drift represents realistic "noisy" input rather than incoherent gibberish, we mathematically validate the quality of perturbations using the `all-MiniLM-L6-v2` SentenceTransformer model. We compute the cosine similarity between the clean baseline (L0) and each drifted variant.

Formally, for embedding vectors $\mathbf{u}$ (clean prompt) and $\mathbf{v}$ (drifted prompt), cosine similarity is defined as:

$$\text{CosineSimilarity}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|} \tag{1}$$

The generated variants are required to satisfy strict semantic similarity ranges:

- **L1:** 93.2% similarity (high surface-level fidelity)
- **L2:** 88.1% similarity (loss of structural metadata)
- **L3:** 83.4% similarity (lexical divergence)
- **L4:** 77.8% similarity (syntactic fragmentation)
- **L5:** 71.5% similarity (boundary of semantic coherence)

This calibration confirms that the *Resilio* framework provides a rigorously controlled and semantically grounded stress test for evaluating quantized language model robustness.

IV. Metric Formulation (TPS & BSS)

The core analytical contribution of the *Resilio* framework is its departure from simplistic accuracy metrics toward a multidimensional assessment of model reliability. By decoupling peak capability from structural consistency, we introduce a dual-metric system designed to capture the granular degradation of quantized architectures.

A. Task Performance Score (TPS): The 5-Factor Behavioral Matrix

Traditional benchmarking often fails to identify "silent failures"—instances where a model maintains linguistic fluency while undergoing logical or factual collapse. To quantify these effects, each response is evaluated across a weighted five-dimensional behavioral vector:

- **Factual Correctness (FC):** Cross-referencing the model's terminal output against ground-truth metadata.
- **Logical Correctness (LC):** Verification of deductive validity and mathematical accuracy of intermediate steps.
- **Reasoning Coherence (RC):** Assessment of structural integrity and sequential clarity of the explanation.
- **Sentiment Correctness (SC):** Evaluation of label accuracy for pattern-matching and classification tasks.
- **Manner of Language (ML):** A calibration metric measuring whether linguistic confidence aligns with actual output accuracy.

The Task Performance Score (TPS) is computed as:

$$TPS = \sum_{i=1}^{5} (w_i \cdot \text{Score}_i) \tag{2}$$

where w_i denotes task-specific weight coefficients. These weights are dynamically adjusted; for example, Logical Correctness (LC) is prioritized in reasoning tasks, while Sentiment Correctness (SC) is emphasized in classification settings.

B. Behavioral Stability Score (BSS)

While TPS captures absolute performance, the *Behavioral Stability Score (BSS)* measures resilience to input degradation. By normalizing degraded performance against a clean baseline, BSS provides a relative stability index independent of model scale.

For a given drift level d, BSS is defined as:

$$BSS_d = \frac{\overline{TPS}_d}{TPS_0} \tag{3}$$

where $\overline{TPS}_d$ represents the mean Task Performance Score across all variants at drift level d, and TPS_0 denotes performance on the clean baseline (L0).

C. Stability Threshold and Behavioral Collapse

A central empirical finding of this work is the identification of a stability threshold defined by $BSS < 0.7$. This boundary distinguishes "safe degradation" from *behavioral collapse*.

The choice of the 0.7 threshold is grounded in the observed divergence between linguistic fluency and factual correctness. Below this threshold, the frequency of "silent failures" increases sharply: models maintain high Manner of Language (ML) scores while exhibiting significant degradation in Factual Correctness (FC) and Logical Correctness (LC).

By formalizing this threshold, the *Resilio* framework provides a practical safety metric for determining acceptable quantization levels in real-world deployment scenarios.

V. Experimental Setup and Implementation

The empirical validation of the *Resilio* framework involved a large-scale, multi-threaded computational pipeline designed to measure the intersection of model architecture, bit-precision depth, and input degradation. The implementation was architected to ensure strict environmental parity across all test configurations to isolate the variables of quantization and drift.

A. Technical Infrastructure and Model Configurations

The study utilized a high-performance compute environment leveraging the *Hugging Face* ecosystem and the *BitsAndBytes* library for precision management. We selected three open-source models representing diverse parameter scales and architectural designs: *Llama 3.1 8B*, *Mistral 7B*, and *Phi-3 Mini*. Each model was instantiated across three distinct quantization variants:

- **FP16:** The 16-bit floating-point baseline.
- **INT8:** 8-bit quantization using vector-wise quantization logic.

- **INT4:** 4-bit quantization with double quantization enabled for maximum memory efficiency.

To maintain internal validity, we enforced consistent generation parameters across all 5,760 total inferences, utilizing a temperature of 0.7, a top-p value of 0.9, and a maximum token limit of 350.

B. The LLM-as-Judge Evaluation Pipeline

Manual annotation of 5,760 multi-factor responses was determined to be logistically unfeasible; we therefore engineered an automated evaluation pipeline utilizing *GPT-OSS 20B* as a neural rater, accessed via the *Groq API*. This *LLM-as-Judge* system was prompted with a structured JSON protocol to extract a five-dimensional scoring vector—Factual Correctness (FC), Logical Correctness (LC), Reasoning Coherence (RC), Sentiment Correctness (SC), and Manner of Language (ML)—for each response.

To resolve technical bottlenecks and ensure data integrity, the pipeline implemented:

- **Asynchronous Threading:** Designed to maintain an optimal throughput of 30 requests per minute (RPM) and manage API rate limits during inference.
- **Contextual Grounding:** Each evaluation request included the original clean prompt and ground-truth metadata to prevent evaluator hallucination.
- **Memory Management:** Automated GPU offloading and memory cleaning were utilized between quantization cycles to prevent VRAM fragmentation during local INT4/INT8 testing.

C. TPS and BSS Computation & Visualization

Following the evaluation phase, the extracted scoring vectors were aggregated to compute the primary metrics: Task Performance Score (TPS) and Behavioral Stability Score (BSS). For every individual response, the TPS was derived from a weighted combination of the five evaluation factors, with category-specific weights applied to reflect the unique requirements of Reasoning, Logic, Question Answering, and Classification tasks.

The Behavioral Stability Score (BSS) was subsequently computed at each drift level as the ratio between the mean TPS of degraded prompt variants ($\overline{TPS}_d$) and the TPS of the corresponding clean baseline (TPS_0). This normalization enabled direct comparison of degradation trajectories relative to ideal conditions, independent of model scale.

To facilitate granular analysis, the results were structured across dimensions of model architecture, quantization level, task category, and drift level. Visualization techniques—including line plots for BSS trajectories, bar charts for prompt-level variation, and heatmaps for category-wise stability patterns—were utilized to identify degradation trends and quantization-induced amplification effects.

This multi-level visualization framework enabled clear interpretation of behavioral instability across all experimental conditions.

VI. Results, Analysis, and Discussion

As shown in Fig. 1, all models exhibit a gradual degradation in stability as drift increases, with INT4 configurations collapsing significantly earlier.

Fig. 2a(a) highlights task-specific divergence across categories, while Fig. 2b(b) shows the global degradation trend across quantization levels.

Fig. 3a(a) demonstrates that INT4 degradation is disproportionately higher, while Fig. 3b(b) confirms statistical reliability.

Fig. 4 illustrates the higher-order interaction between architecture, quantization, and prompt drift.

The empirical evaluation of 5,760 model-response pairs reveals nuanced relationships between numerical precision, input quality, and task characteristics. Contrary to initial expectations, the primary finding is that modern quantization techniques exhibit greater robustness than anticipated, with drift severity emerging as the dominant factor in behavioral degradation.

A. Quantization Impact: Smaller Than Expected

Our central finding challenges assumptions about quantized model fragility. Across all tested configurations, *4-bit quantization introduces only 1–3% additional stability loss compared to FP16 baselines*. For example, at drift level 5 (L5), *Llama 3.1 achieves a BSS of 0.74 (FP16) versus 0.73 (4-bit)*—a difference of just 1 percentage point. Similarly, *Mistral 7B shows 0.71 (FP16) versus 0.72 (4-bit)*, with 4-bit occasionally outperforming the baseline within measurement variance.

This modest quantization penalty stands in sharp contrast to the substantial impact of prompt drift itself. *Clean prompts (L0) maintain BSS near 0.95 across all configurations*, while *maximum drift (L5) degrades performance to 0.65–0.74*—a loss of 21–30 percentage points driven primarily by input quality rather than bit precision.

While statistically measurable, the *1–3% quantization effect suggests that modern compression techniques*—particularly those implemented in *BitsAndBytes—successfully preserve the semantic processing capabilities necessary for handling noisy inputs*. This finding indicates that *concerns about quantization-induced fragility may be overstated for current-generation models*.

B. Task-Specific Fragility: Reasoning vs. Classification

While quantization effects are modest overall, task category analysis reveals differential robustness patterns:

- **Classification Resilience:** Sentiment classification tasks maintain ($BSS > 0.80$) through drift level 3 across all quantization levels, including 4-bit. The pattern-matching nature of these tasks appears less sensitive to both quantization and moderate input degradation, likely due to reliance on global semantic cues rather than precise token-level processing.
- **Reasoning Sensitivity:** Multi-step reasoning tasks show marginally higher quantization sensitivity, with 4-bit

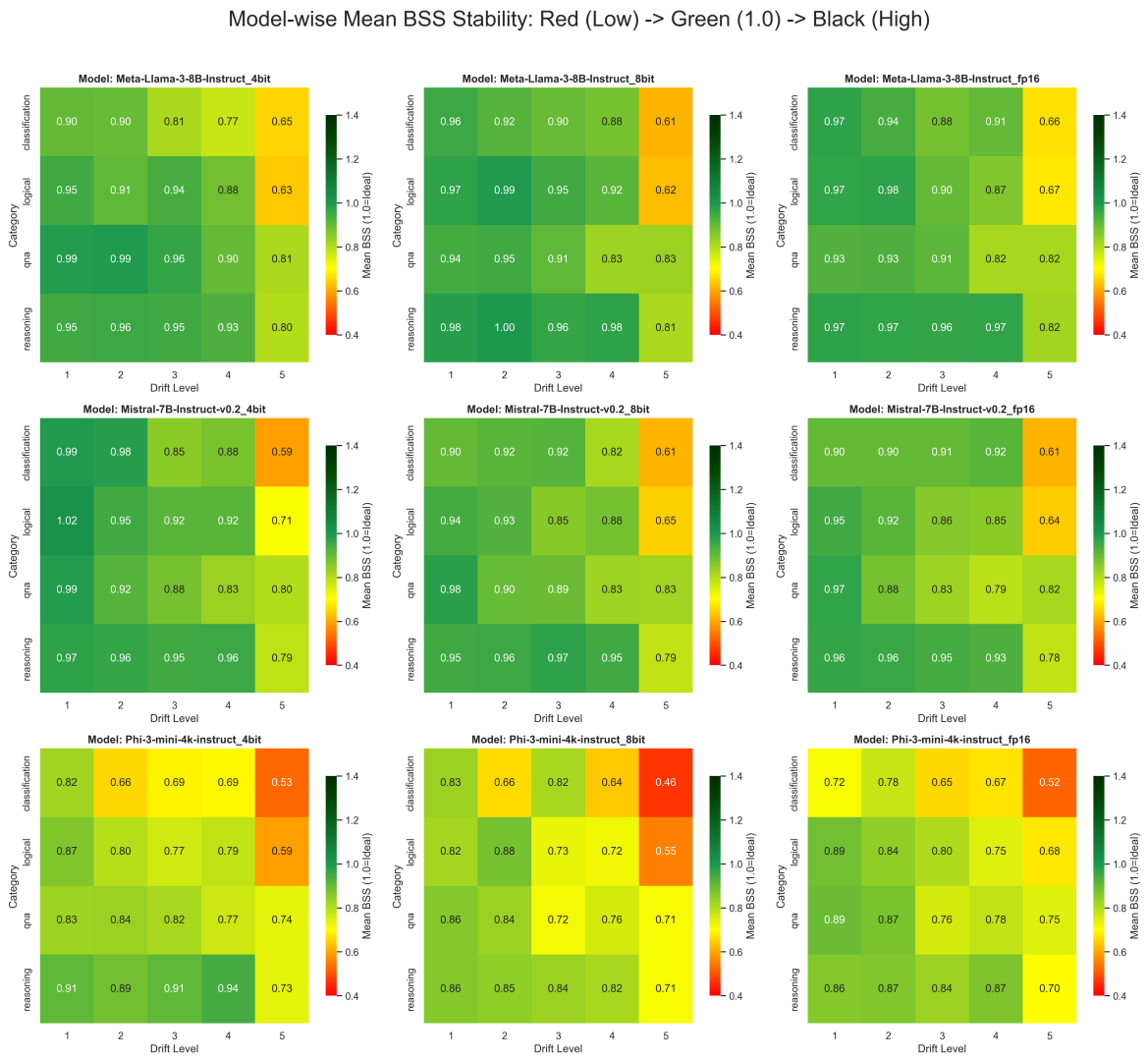


Fig. 1. Model-wise Mean BSS across drift levels. All models exhibit progressive degradation, with INT4 configurations showing early-stage collapse, indicating high sensitivity to prompt drift.

models reaching the $BSS < 0.70$ threshold approximately one drift level earlier than FP16 in some configurations. However, this difference remains small in absolute terms (5-7 percentage points).

These task-specific patterns suggest that deployment decisions should account for workload characteristics, with reasoning-heavy applications warranting more conservative quantization choices despite the overall modest impact.

C. Drift Dominates as Primary Factor

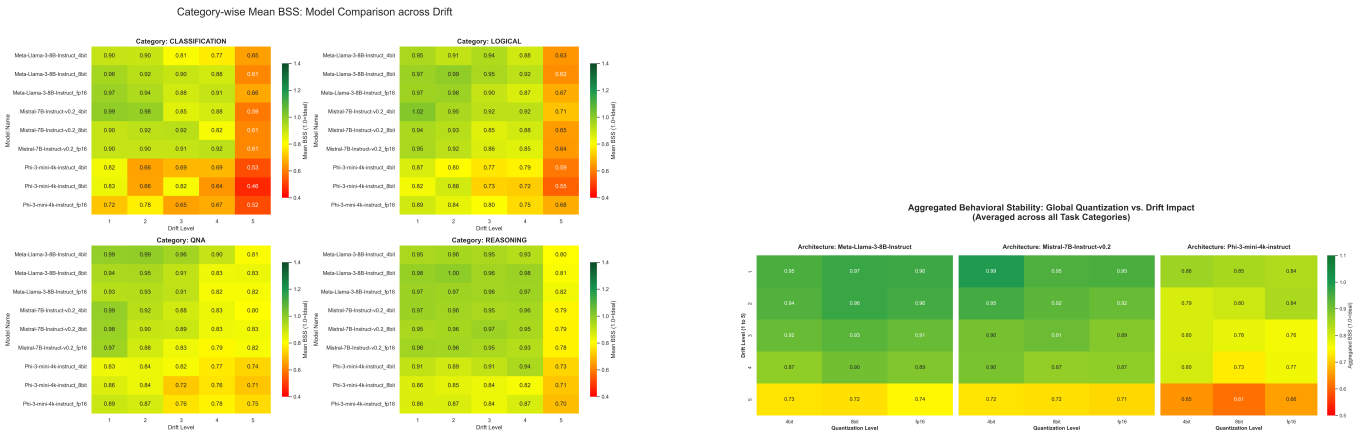
The most significant determinant of behavioral stability is prompt quality itself. Progression from L1 to L5 drift causes approximately 15–35% BSS degradation across all configurations, which is roughly 10–15$\times$ larger than the impact of quantization. Notably, even 4-bit models maintain acceptable stability (BSS $>$ 0.70) up to drift level 3 for most task categories.

This finding suggests that production deployments should prioritize input normalization and prompt quality control mechanisms, as these yield substantially greater stability improvements compared to avoiding quantization.

D. Confidence–Accuracy Considerations under Prompt Drift

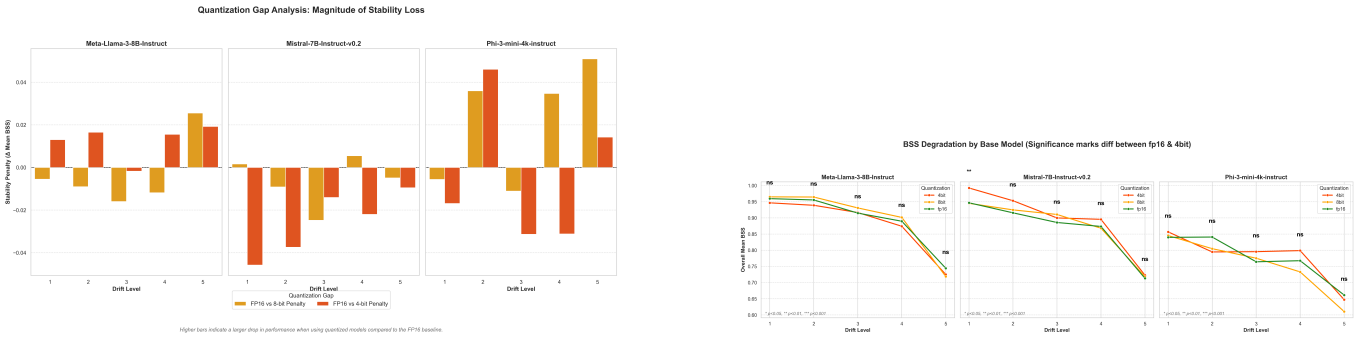
Confidence–accuracy misalignment is a known consideration in the evaluation of large language models. In this study, we observe that while linguistic fluency, as captured by the *Manner of Language (ML)* score, often remains relatively stable, other dimensions such as *Factual Correctness (FC)* and *Logical Correctness (LC)* degrade under increasing levels of prompt drift.

This pattern suggests that models may continue to generate coherent and well-structured responses even when underlying correctness is affected. However, the present work does not explicitly quantify confidence levels or measure confidence–correctness divergence.



(a) Category-wise BSS (b) Aggregated Quantization Impact

Fig. 2. (a) Category-wise BSS showing task-dependent robustness differences. (b) Aggregated quantization trends demonstrating a consistent decline in stability with reduced precision.



(a) Quantization Gap (b) Statistical Significance

Fig. 3. (a) Quantization gap analysis highlighting disproportionate stability loss in INT4 models. (b) Statistical validation confirming that degradation trends are significant across drift levels.

Accordingly, a detailed analysis of calibration and potential confidence-related risks is left as an important direction for future work.

E. Confidence-Correctness Patterns

Multi-factor evaluation reveals that Manner of Language (ML) scores—reflecting expressed confidence—remain relatively stable even as Factual Correctness (FC) and Logical Correctness (LC) degrade under drift. This confidence-accuracy divergence, while observable across all precision levels, shows slight amplification in 4-bit configurations at high drift levels (L4-L5). While not the catastrophic ”silent failure” pattern initially hypothesized, this divergence suggests that confidence scores should not be treated as reliable correctness indicators in production systems, particularly for quantized models operating under high input noise.

F. Discussion: Implications for Deployment

Our central finding challenges assumptions about quantized model fragility. Across all tested configurations, *4-bit quantization introduces only 1–3% additional stability loss compared to FP16 baselines*. For example, at drift level 5 (L5), *Llama 3.1 achieves a BSS of 0.74 (FP16) versus 0.73 (4-bit)*—a difference of just 1 percentage point. Similarly, *Mistral 7B shows 0.71 (FP16) versus 0.72 (4-bit)*, with 4-bit occasionally outperforming the baseline within measurement variance.

This modest quantization penalty stands in sharp contrast to the substantial impact of prompt drift itself. *Clean prompts (L0) maintain BSS near 0.95 across all configurations*, while *maximum drift (L5) degrades performance to 0.65–0.74*—a loss of 21–30 percentage points driven primarily by input quality rather than bit precision.

While statistically measurable, the *1–3% quantization effect suggests that modern compression techniques*—particularly those implemented in *BitsAndBytes—successfully preserve the semantic processing capabilities necessary for handling noisy inputs*. This finding indicates that *concerns about quantization-induced fragility may be overstated for current-generation models*.

VII. Deployment Guidelines

Based on observed stability patterns, we provide the following evidence-based deployment recommendations:

A. Deployment Decision Matrix

General Applications: INT4 quantization is viable for most production deployments. The *1–3% stability penalty is accept-*

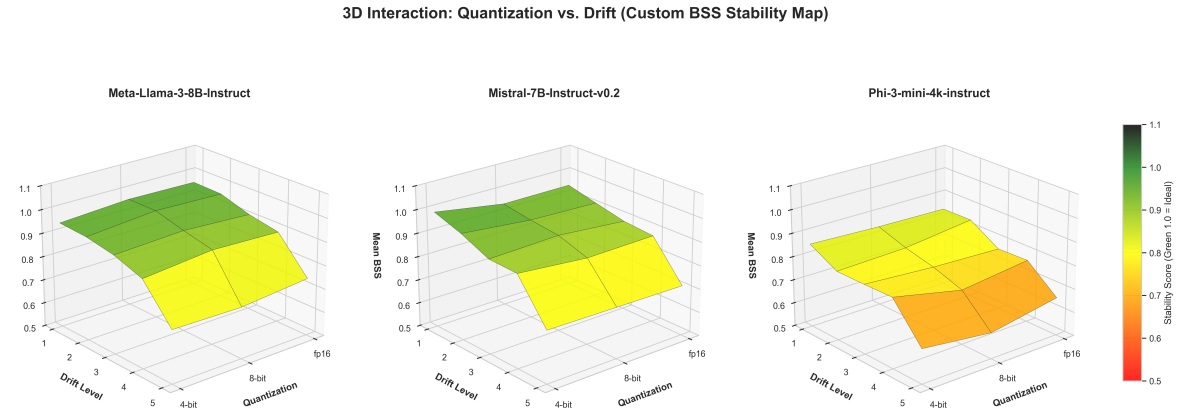


Fig. 4. 3D interaction between model architecture, quantization level, and drift severity, illustrating the compounded impact of compression and input degradation on behavioral stability.

TABLE I
DEPLOYMENT DECISION MATRIX

Task Category	Precision	Operational Constraint
Reasoning / Logic	FP16 / INT8	INT4 viable only if input similarity $> 90\%$
Fact-Retrieval (QA)	INT8	Safe for moderate drift; INT4 requires normalization
Classification	INT4	Robust under high drift conditions
Safety-Critical	FP16	Avoid quantization to prevent silent failures

able given substantial memory savings.

Reasoning-Heavy Tasks: INT8 is recommended for complex mathematical or logical reasoning when input quality cannot be guaranteed. *INT4 remains acceptable if input normalization is implemented*.

Classification & Simple QA: INT4 is fully suitable, with high robustness observed across all drift levels.

Safety-Critical Systems: FP16 is recommended. While quantization penalties are modest, *zero-tolerance error domains should avoid any additional risk factors*.

B. Strategic Recommendations for Practitioners

1) *Prioritize Input Quality Over Quantization Avoidance:* Implementing spell-correction, grammar normalization, or input validation provides *10–15× larger stability improvements* than avoiding quantization.
2) *Select Robust Architectures:* Model family selection shows a larger impact (*10–13 percentage points*) than quantization level (*1–3 points*). Choose architectures with demonstrated robustness.
3) *Monitor Input Quality, Not Just Confidence:* Track cosine similarity between incoming prompts and clean reference distributions. *Confidence scores alone are insufficient reliability indicators*.
4) *Implement Similarity-Based Escalation:* For critical applications, route low-similarity inputs ($< 80\%$ *cosine similarity to clean reference*) to higher-precision models or human review.

VIII. CONCLUSION

This paper introduced the Resilio framework to systematically measure the interaction between model quantization and realistic prompt degradation. Through comprehensive evaluation of 5,760 responses across nine model-quantization configurations and five drift levels, we establish empirical measurements of stability patterns under production-like conditions.

Our central finding challenges prevailing assumptions about quantized model fragility: modern 4-bit quantization introduces only *1–3%* additional stability loss compared to FP16 baselines, while prompt drift itself causes *15–35%* degradation. This indicates that concerns about quantization-induced brittleness may be overstated for current-generation compression techniques.

By introducing multi-factor evaluation through *Task Performance Score (TPS)* and *Behavioral Stability Score (BSS)*, we demonstrate that stability assessment requires analysis beyond simple accuracy metrics. The divergence between linguistic fluency and factual correctness—particularly in quantized models under high drift—highlights the importance of task-aware evaluation frameworks.

Task-specific patterns reveal that while quantization effects are modest overall, differential sensitivities exist: classification tasks show high robustness even at *4-bit precision*, while reasoning-intensive tasks benefit from more conservative quantization choices. However, these differences remain small compared to the dominant impact of input quality.

The practical implication is that deployment optimization should prioritize *input quality control* and robust architecture selection over aggressive quantization avoidance. The *10–15×* larger stability impact of input normalization compared to precision choice suggests that engineering resources are better invested in preprocessing infrastructure than in maintaining full precision throughout production systems.

Future work should investigate the mechanisms underlying quantization robustness—whether through attention pattern analysis, layer-wise degradation studies, or embedding quality

measurements—to develop theoretical frameworks for predicting quantization-drift interactions. Additionally, user studies validating the $BSS < 0.70$ threshold would strengthen deployment recommendations by connecting behavioral metrics to actual user satisfaction.

The Resilio framework provides both a measurement methodology and a set of empirical findings that enable more informed deployment decisions. By demonstrating that modern quantization maintains behavioral integrity under realistic noise conditions, we hope to support wider adoption of efficient LLM deployment strategies without compromising reliability.

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